

What Is Learning?

Supervised · Unsupervised · Reinforcement

Topic 1 of Core ML Concepts

 Labeled examples

 Hidden structure

 Rewards over time

A Working Definition of Learning

A program learns when experience improves its performance on a task.

E

Experience

Examples, interactions, or observations

T

Task

Predict, discover, or decide

P

Performance

A measurable objective or reward

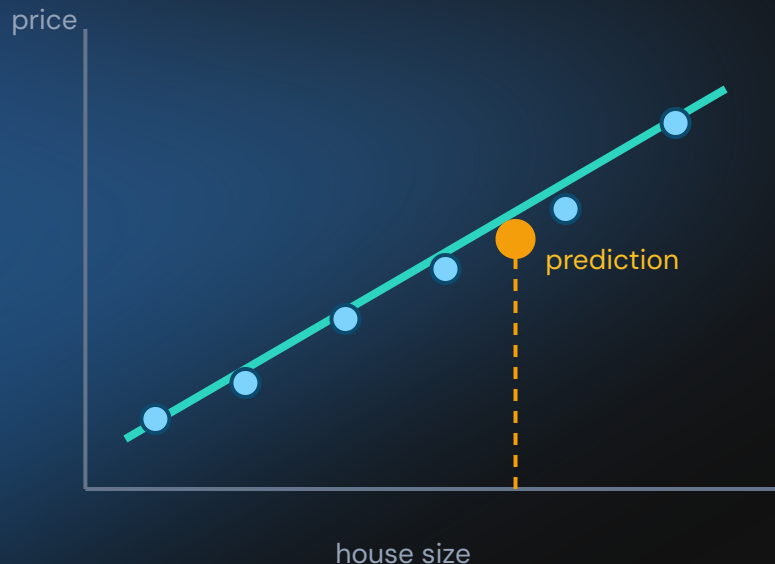
$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(f_{\theta}(x), y)]$$

Data supplies experience · loss measures performance · optimization changes the model

Supervised Learning

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n, \quad f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$$

- Every input arrives with a **known target**
- Learn a mapping that generalizes to unseen inputs
- **Regression**: continuous targets, such as price
- **Classification**: discrete targets, such as spam/not-spam



Supervised Learning — A Tiny Regressor

```
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.array([[1200], [1500], [1800], [2200]]) # square feet
y = np.array([300000, 340000, 400000, 480000]) # sale price

model = LinearRegression().fit(X, y)
print(model.predict([[2000]]))
```

Examples

labeled houses

Fit

learn slope + intercept

Generalize

price a new house

Unsupervised Learning

$$\mathcal{D} = \{x_i\}_{i=1}^n \quad \text{no labels}$$

- Discover structure rather than predict known targets
- **Clustering**: find groups
- **Dimensionality reduction**: compress while preserving structure
- **Density estimation**: model where data is likely to occur
- Evaluation is task-dependent because there may be no single “correct” answer



Unsupervised Learning — Discovering Groups

```
from sklearn.cluster import KMeans
import numpy as np

X = np.array([[1, 2], [1.5, 1.8], [1, 0.6],
              [8, 8], [9, 11], [8.5, 10]])

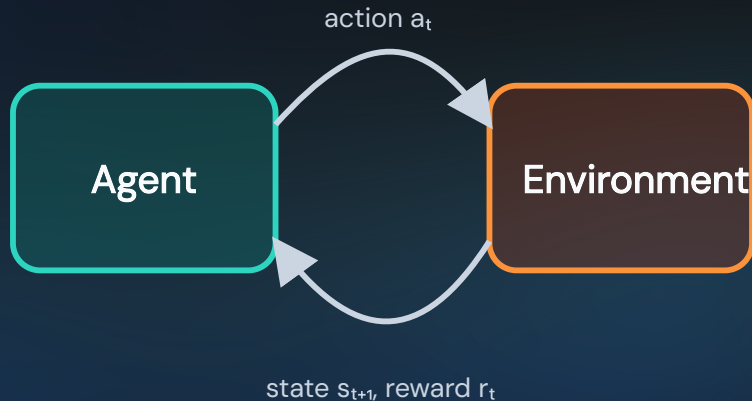
model = KMeans(n_clusters=2, n_init='auto', random_state=0).fit(X)
print(model.labels_)          # assignments discovered from geometry
print(model.cluster_centers_)
```

The algorithm invents cluster IDs; humans still decide whether the grouping is useful.

Reinforcement Learning

- An **agent** chooses actions inside an **environment**
- The environment returns a new state and a reward
- Learn a **policy** that maximizes cumulative reward
- Feedback is evaluative, often delayed—not a labeled correct action

$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r_t \mid \pi \right]$$



Between the Big Three

Semi-supervised

Combine a **small labeled set** with a much larger pool of unlabeled data.



Self-supervised

Create targets from the data itself—such as predicting a **masked word** from context.

The model reads the [MASK] .

Modern systems often combine paradigms: **self-supervised pretraining** → **supervised fine-tuning**

Comparing the Paradigms

	Supervised	Unsupervised	Reinforcement
Data	(x, y) pairs	x only	interaction stream
Feedback	direct labels	none inherent	rewards
Goal	predict	discover structure	choose actions
Example	linear regression	k-means	Q-learning
Use case	price prediction	customer segments	robot control

Where This Course Focuses

Most

Supervised learning

Regression, classification, trees, SVMs, and neural networks

A unit

Unsupervised

Clustering, PCA, and mixture models



Reinforcement learning: recognize the structure; a deep treatment belongs in its own course.

Next: **Bias–Variance Tradeoff** — why fitting the data is not the same as learning the pattern.